

Informe de experimento: react_nogate

Generado el 2026-09-21 18:12 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio react_nogate (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f ]
↪ low-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: react_nogate
- Casos: case14
- Requests: generadas, N=40 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=10, parameterized=10, multistep=10, ambiguous=10
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 13:31
- Costo total: 1.3219 USD (40 items; costo medio 0.03305 USD/item)
- Items totales: 40; ok=40; con error=0; tokens totales=458729
- Solved global: 60.0% (24/40)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo. **Esta reconstrucción es ilustrativa, no histórica:** usa el código de llm/prompts.py de HOY, así que si el texto del prompt cambió después de que esta corrida se generó, lo que se ve aquí no es lo que el modelo recibió. El registro histórico real es el hash por fila (system_prompt_hash, ver sección 1 / benchmarks/experiment_report.py::section_what_ran), fijado en el momento de la corrida y nunca recalculado.

Método react_nogate

System prompt (SYSTEM_PROMPT_EN) (reconstrucción con el código actual, no necesariamente histórica) — 1557 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given, unless the request itself states a
  ↪ different indexing convention for that identifier (e.g. "(0-based)"), in which case convert it to the MATPOWER 1-based id before
  ↪ calling any tool.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

User prompt (primer item) = request_text -67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

3. Resultados agregados

Columnas del protocolo (medias por método $\times$ caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfaqent_definicion` **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u., solo sobre peticiones con formulación exacta -toda fila con herramientas que llega al solver con la formulación correcta da exactamente cero, ver benchmarks/scoring.py) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (por respuesta, no por número: fracción de respuestas donde todo número es trazable a una salida de tool Y viene del solve posterior al último cambio de red) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Escalated y Wrong (silent). Escalated y Wrong (silent) junto con solved_autonomously_rate (no la columna Solved de arriba, que puede solaparse con Escalated -una abstención de verificación puede caer sobre un item cuyo cálculo de fondo sí era correcto) suman 100 %: dónde termina cada petición -resuelta sola, entregada a una persona, o mal contestada sin aviso; solo cuenta como Escalated una arquitectura con mecanismo de traspaso real, ver benchmarks.scoring.escalation_check) y \$ (costo total USD). Porcentajes en %.

Por método $\times$ caso

Method	Case N	Form.	V_MAE solved	V_MAE all	B_mean solved	B_mean all	Solved	Wrong-Escalated	Wrong-Flagged	Traceable	Time Token(s)	Claim	Wrong-Escalated	Wrong (silent)	\$
react_nogate	case14	62.5 (25/40)	0	1.13e-04	-	0	60.0 (24/40)	0.0 (0/40)	40.0	90.0	114686.49	-	0.0 (0/40)	40.0 (16/40)	1.3219

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
react_nogate	80.0 (8/10)	90.0 (9/10)	40.0 (4/10)	40.0 (4/10)

Solved por dificultad (%, items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
react_nogate	80.0 (8/10)	90.0 (9/10)	30.0 (3/10)	40.0 (4/10)

Tipos de error de formulación (conteo de items)

Method	extra_arg	changes_result	ok
react_nogate	15		25

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

Método react_nogate

Ejemplo: resuelto (solved=True)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 5930 chars más]
gate: PASS (reasons=, mismatch=7.85e-09 MW)
--- Ronda 3 ---
LLM: 1. One-sentence summary
I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the post-outage power flow converged.

2. Key numbers
- Total load: 253.846 MW
- Total generation: 266.510 MW
- Total losses: 12.665 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if any [504 chars más]
--- Respuesta final ---
1. One-sentence summary
I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the post-outage power flow converged.

2. Key numbers
- Total load: 253.846 MW
- Total generation: 266.510 MW
- Total losses: 12.665 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if any
- Voltage violations:
- Bus 1: 1.060 pu, overvoltage
- Bus 6: 1.070 pu, overvoltage
- Bus 7: 1.057 pu, overvoltage
- Bus 8: 1.090 pu, overvoltage
- Bus 11: 1.064 pu, overvoltage
- Bus 12: 1.054 pu, overvoltage
- Thermal violations: none
- Disconnected line status:
- Line 10-11 flow is 0.0 MW / 0.0 MVar as expected

4. Suggested next step
If you want, I can now check the most loaded branch, run an N-1 contingency analysis, or suggest remedial actions for the voltage violati ...
↳ [4 chars más]
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_and_answer_ok); flags=-; error=-; tokens=5350;
↳ cost_usd=0.0173; wall=7.0 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-ambiguous-011-s0, case14, difficulty=ambiguous, seed=1971893210): Load case14 and set the active load at bus 6 to 14,700 kw.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 6, "p_mw": 14.7})
executed_calls: load_case({"case_name": "case14"}) -> get_status({}) -> modify_load({"bus_id": 6, "p_mw": 14.7, "q_mvar": 11.2})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 263.13580543670497,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: get_status({})
output: {"active_case": "case14", "network_info": {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11,
↳ "total_load_mw": 263.13580543670497, "total_gen_capacity_mw": 772.4000000000001}, "has_last_result": false}
gate: n/a (no es un resultado de flujo)
--- Ronda 3 ---
tool: modify_load({"bus_id": 6, "p_mw": 14.7, "q_mvar": 11.2})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.236323374534588, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 6297 chars más]
gate: PASS (reasons=, mismatch=6.47e-09 MW)
--- Ronda 4 ---
LLM: 1. Summary: I loaded IEEE case14 and set bus 6 active load to 14,700 kW (interpreted as 14.7 MW), and the power flow converged.

2. Key numbers:
- Total load: 267.145 MW
```

```
- Total generation: 281.819 MW
- Total losses: 14.674 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if any:
- Voltage v ...[257 chars más]
--- Respuesta final ---

1. Summary: I loaded IEEE case14 and set bus 6 active load to 14,700 kW (interpreted as 14.7 MW), and the power flow converged.

2. Key numbers:
- Total load: 267.145 MW
- Total generation: 281.819 MW
- Total losses: 14.674 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if any:
- Voltage violations: overvoltage at buses 1, 6, 7, 8, 9, 10, 11, and 12
- Thermal violations: none

4. Suggested next step:
- If you want, I can also adjust the reactive load at bus 6, check the most loaded branch, or run a contingency analysis on this modified
  ↳ case.
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=9.52e-17; F_MAE=0.00; faithful=100.0 %; solved=False (formulation); flags=-;
↳ error=-; tokens=6540; cost usd=0.0195; wall=5.4 s
```

```
request (case14-multistep-030-s0, case14, difficulty=multistep, seed=537958268): Load case14, set the active load at bus 14 to 17.2 MW,  
→ disconnect the line between bus 1 and bus 5, reconnect the line between bus 1 and bus 5, and finally run an N-1 contingency analysis  
→ ranked by min voltage and report the 8 worst outages.  
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 14, "p_mw": 17.2}) -> disconnect_line({"from_bus": 1,  
→ "to_bus": 5}) -> reconnect_line({"from_bus": 1, "to_bus": 5}) -> run_n1_contingency({"top_k": 8, "criteria": "min_voltage"})  
executed_calls: load_case({"case_name": "case14"}) -> get_status({}) -> modify_load({"bus_id": 14, "p_mw": 17.2, "q_mvar": 0}) ->  
→ disconnect_line({"from_bus": 1, "to_bus": 5}) -> reconnect_line({"from_bus": 1, "to_bus": 5}) -> run_n1_contingency({"top_k": 8,  
→ "criteria": "min_voltage", "max_candidates": 0})  
fuente de la traza: archivo de traza completo  
--- Ronda 1 ---  
tool: load_case({"case_name": "case14"})  
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 271.9322468585065,  
→ "total_gen_capacity_mw": 772.4000000000001}  
gate: n/a (no es un resultado de flujo)  
--- Ronda 2 ---  
tool: get_status({})  
output: {"active_case": "case14", "network_info": {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11,  
→ "total_load_mw": 271.9322468585065, "total_gen_capacity_mw": 772.4000000000001}, "has_last_result": false}  
gate: n/a (no es un resultado de flujo)  
--- Ronda 3 ---  
tool: modify_load({"bus_id": 14, "p_mw": 17.2, "q_mvar": 0})  
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,  
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.4514286734574515, "is_violation": false,  
→ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "v ...[6408 chars más]  
gate: PASS (reasons=-, mismatch=5.92e-09 MW)  
tool: disconnect_line({"from_bus": 1, "to_bus": 5})  
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,  
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -8.382663773008685, "is_violation": false,  
→ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0099999 ...[6332 chars más]  
gate: PASS (reasons=-, mismatch=2.66e-08 MW)  
tool: reconnect_line({"from_bus": 1, "to_bus": 5})  
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,  
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.4514286734574515, "is_violation": false,  
→ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "v ...[6408 chars más]  
gate: PASS (reasons=-, mismatch=5.92e-09 MW)  
--- Ronda 4 ---  
tool: run_n1_contingency({"top_k": 8, "criteria": "min_voltage", "max_candidates": 0})  
output: {"plot_type": "n1_ranking":  
→ "{ \"data\": [{ \"cliponaxis\": false, \"hoverinfo\": \"text\", \"hovertext\": [ \\\"\\u003cb\\\"\\u003eOutage\\\"\\u003c\\\"\\u002fb\\\"\\u003e: 1-5  
→ (line)\\\"\\u003cbr\\\"\\u003e\\\"\\u003cb\\\"\\u003eConverged\\\"\\u003c\\\"\\u002fb\\\"\\u003e: True\\\"\\u003cbr\\\"\\u003e\\\"\\u003cb\\\"\\u003eVoltage  
→ violations\\\"\\u003c\\\"\\u0 ...[19527 chars más]  
gate: n/a (no es un resultado de flujo)  
--- Ronda 5 ---  
LLM: 1. One-sentence summary
```

2. Key numbers

1. One-sentence summary

2. Key numbers

From the final restored-network power flow after reconnecting line 1-5:

- Total load: 275.141429955464 MW
- Total generation: 290.6125020948048 MW
- Total losses: 15.471072145264227 MW
- Voltage range: 1.043390055906286 pu to 1.0899999999999999 pu

3. Violations, if any

Base case after the load change and line reconnection:

- Voltage violations: 9 overvoltage violations at buses 1, 6, 7, 8, 9, 10, 11, 12, and 13
- Thermal violations: none

Requested N-1 results ranked by minimum voltage, 8 worst outages:

1. Outage 1-5 (line) ...[1792 chars más]

--- Scoring ---

formulation=extra_arg_changes_result (exact=False); V_MAE=1.62e-03; F_MAE=0.08; faithful=100.0 %; solved=False (formulation);

→ flags=stale_state; error=-; tokens=30617; cost_usd=0.0890; wall=11.4 s

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- $V_MAE \geq 1e-4$ p.u. en 1 filas método×caso con solver: react_nogate/case14 (1.13e-04).
- El gate verificó 58 resultados de flujo, 0 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved de react_nogate: 60.0 % (24/40).
- Formulación exacta agregada sobre métodos con tools: 62.5 % (25/40).
- Formulación por dificultad: mejor parameterized 90.0 % (9/10), peor multistep 40.0 % (4/10).
- Errores de formulación más frecuentes: extra_arg_changes_result=15.
- Ningún item reclamó éxito sobre un fallo (claimed_success_on_failure=0).
- 1 de 26 items con mutación de red reportaron estado obsoleto (stale_state).
- Trazabilidad de números (Faith.) media con tools: 99.0 %.
- Costo total 1.3219 USD; el método más caro fue react_nogate con 1.3219 USD (100 %).
- Métodos con tools: 2.83 tool calls medias por item (máximo 6).

6. Archivos

Rutas absolutas bajo react_nogate.report.rescored.json (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/report.json [leído:
↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard.json
↳ ids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard.csv
↳ ds-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard.md
↳ s-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard_per_case.json
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard.rescored.json
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard.rescored.json
scoreboard_per_case.rescored.json:
↳ /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/scoreboard_per_case.rescored.json
traces (40 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/traces
↳ artGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/react_nogate/traces
```